

Central Bank Information Effects in FOMC Announcements:

Evidence from the U.S. Treasury Yield Curve

Bachelor's Thesis

Eric Oehler

23-621-477

Supervised by: Prof. Dr. Martin Wolf

Submission Date: 19 May 2026

Abstract

This paper contributes to the debate on Central Bank Information effects. To this end, it examines whether Federal Open Market Committee announcements with positive co-movement between monetary policy surprises and stock prices exhibit systematically different U.S. Treasury yield curve responses than announcements with negative co-movement. It relies on asset price changes in 30-minute event windows around Federal Open Market Committee announcements and applies principal component analysis to summarize changes in Treasury yields by the level, slope, and curvature factors. The main result is that contractionary surprises flatten the Treasury yield curve, and that this flattening is attenuated during announcements with positive co-movement. While this pattern is consistent with information-related transmission channels of monetary policy, it does not provide conclusive evidence for Central Bank Information effects, as alternative explanations cannot be ruled out.

Contents

List of Tables **III**

List of Figures **III**

List of Abbreviations **IV**

1 Introduction **1**

2 Literature **3**

2.1 Monetary Policy Surprises and High-Frequency Identification 3

2.2 Transmission to the Stock Market 5

2.3 Transmission to the U.S. Treasury Market 7

2.4 Implications 9

3 Empirical Strategy **11**

3.1 Data and Sample Construction 11

3.2 Measurement of Monetary Policy Surprises 12

3.3 Event Classification 13

3.4 Treasury Yield Curve Factor Construction 14

3.5 Regression Specification 17

4 Results **19**

4.1 Descriptive Evidence 19

4.2 Baseline Results 22

4.3 Main Results: Event Heterogeneity 23

4.4 Sensitivity Analysis 24

4.5 Economic Interpretation 26

5 Conclusion **28**

References **29**

Appendix **32**

Declaration of Authorship **35**

Directory of Auxiliary Aids **36**

List of Tables

1 Orthogonal MPS and Event Classification 19
2 Explained Variance of Principal Components 20
3 Baseline Specification 22
4 Extended Specification 23
A.1 Extended Specification with Conservative (HC3) Standard Errors 32
A.2 Extended Specification with Sample Period 1995-2023 32
A.3 Extended Specification with Sample Period 1997-2023 33
A.4 Extended Specification with Conventional MPS 33
A.5 Extended Specification with Stricter Event Classification 34
A.6 Extended Specification with Treasury Yield Spreads 34
A.7 Use of Supplementary Materials 36

List of Figures

1 Loadings on Principal Components across Maturities 21

List of Abbreviations

CBI Central Bank Information

Fed Federal Reserve System

FOMC Federal Open Market Committee

HC1 Heteroskedasticity-Consistent Standard Errors of Type 1

HC3 Heteroskedasticity-Consistent Standard Errors of Type 3

MPS Monetary Policy Surprises

PCA Principal Component Analysis

SEP Summary of Economic Projections

1 Introduction

Central banks cannot directly determine inflation or real economic outcomes. Instead, they affect economic behavior by influencing interest rates, expectations, and financial conditions. Under standard monetary policy transmission, monetary tightening dampens economic activity. For stock markets this implies that contractionary monetary policy should lead to lower stock prices. However, financial market reactions to monetary policy sometimes differ from these predictions. In particular, there is sometimes positive co-movement between monetary policy surprises (MPS) and stock prices. MPS measure the unanticipated component of monetary policy, typically inferred from high-frequency movements in money market interest rate futures around Federal Open Market Committee (FOMC) announcements. Positive MPS correspond to higher interest rates and accordingly are contractionary surprises. Consequently, stock prices sometimes rise in response to a surprise monetary tightening and fall in response to a surprise monetary easing.

Some authors attribute this puzzle to Central Bank Information (CBI) effects, according to which central banks reveal superior information that goes beyond monetary policy itself (Jarociński & Karadi, 2020). Yet this interpretation is challenged by recent empirical evidence that casts doubt on whether central banks possess information that is systematically superior to that of market participants (Bauer & Swanson, 2023a). The debate matters as it may affect conclusions about monetary non-neutrality (Nakamura & Steinsson, 2018). It also matters to policymakers, since monetary easing could generate pessimism if markets interpret it as a signal that the central bank, based on its superior information, has a weak economic outlook (Bauer & Swanson, 2023b, p. 697).

To contribute to the debate on CBI effects, this paper compares FOMC announcements with positive and negative co-movement between MPS and stock prices. More precisely, it tests whether MPS during FOMC announcements with positive co-movement between MPS and stock prices also exhibit systematically different high-frequency U.S. Treasury yield curve responses. The underlying idea is that if positive co-movement between MPS and stock prices reflects information-related effects, this should not only be visible on the stock market. It should also be reflected in the maturity structure of Treasury yields, because Treasuries with different maturities respond differently to policy action, expected path, and uncertainty components of MPS (Kaminska et al., 2021).

To support the plausibility of a causal interpretation, the analysis uses high-frequency data of asset responses measured within 30-minute event windows around FOMC announcements. After creating a classification indicator based on the sign of co-movement between MPS and stock

prices, Treasury yield curve factors, as summarized by principal components of high-frequency Treasury yield changes across maturities, are regressed on MPS, the classification indicator, and their interaction. The regression results provide some evidence that is consistent with information-related transmission channels of monetary policy during FOMC announcements with positive co-movement between MPS and stock prices.

The remainder of this paper is structured as follows. Chapter 2 reviews the transmission of monetary policy to the stock and Treasury markets to develop important implications for the empirical analysis. Subsequently, Chapter 3 describes the empirical framework and the data used for the analysis. In Chapter 4, the results are presented and discussed. To conclude, Chapter 5 provides a summary of the main results, their limitations, and their implications for future research.

2 Literature

The dual mandate of the Federal Reserve System (Fed) is to maintain a stable price level and achieve maximum employment. Because policymakers must guide economic behavior to achieve these ultimate objectives, the most immediate and direct effects of monetary policy are on asset prices and returns in financial markets (Bernanke & Kuttner, 2005, p. 1221). Thus, the present study concentrates on the transmission of monetary policy to the stock market and the U.S. Treasury market. In this chapter, the relevant literature on how to measure monetary policy and how it is transmitted to the stock and Treasury markets is reviewed. The aim is to provide the theoretical rationale for the empirical analysis.

2.1 Monetary Policy Surprises and High-Frequency Identification

Due to the intention of central banks to influence economic and financial conditions, monetary policy is inherently endogenous, making it difficult for researchers to identify its causal effects on the economy (Rigobon & Sack, 2003). For example, commonly used vector autoregression approaches require assumptions to address the reverse causality between interest rates and economic developments (Swanson, 2011, pp. 8–9). High-frequency identification directly approaches these reverse causality issues by focusing on narrow event windows around monetary policy events.

The high-frequency literature relies on MPS for the estimation of monetary policy effects on financial markets to account for the forward-looking behavior of financial market participants. This is necessary because ignoring the anticipation of monetary policy would bias the estimated effects, as financial markets react primarily to unanticipated monetary policy (Kuttner, 2001). MPS capture this unanticipated part by measuring asset price movements, typically changes in money market interest rate futures, within narrow event windows around monetary policy announcements (Kuttner, 2001). With this approach, the actual effects of monetary policy on financial variables can be estimated more plausibly because the anticipated part of monetary policy is already incorporated into prices before the event window and therefore not reflected in the MPS measure.

Moreover, in forward-looking markets, it is not necessary for the current policy rate to change in order to generate MPS and affect market outcomes. Central banks can also do so through their communication (Gürkaynak et al., 2005). For example, central banks provide information

on the future policy path or the economic outlook, and thereby shape expectations and affect asset prices. MPS thus capture monetary policy that extends beyond the immediate policy rate decision and purge any anticipated part. These characteristics make MPS particularly suitable for the present analysis, where the effects of central bank communication are of central interest.[1]

In the present context, MPS are applied in high-frequency event-study regressions, which exploit asset price movements in narrow event windows around FOMC announcements. These event-study regressions usually take the form of

$$a_t = \beta_0 + \beta_1 \cdot \text{mps}_\text{t} + u_t \,, \tag{1}$$

with a_t denoting an asset response and mps_t representing the MPS measure, both obtained from the same high-frequency event window (Bauer & Swanson, 2023b, p. 111).

Since MPS capture the unanticipated part of monetary policy, this approach promises to identify the causal effects of monetary policy on financial markets (Bauer & Swanson, 2023b, pp. 110–111). The main assumption is that changes in asset prices within the event window occur due to information released as part of the FOMC announcement and not other information (Nakamura & Steinsson, 2018). This assumption is justified, as the short duration of the event window makes it unlikely that any other information, which is correlated with mps_t and a_t, arises (Bauer & Swanson, 2023b, p. 111; Nakamura & Steinsson, 2018, p. 1289). Concerns about omitted variable bias are accordingly mitigated. As mentioned above, reverse causality concerns are also addressed. The reason for this is that FOMC decisions and statements are completed several hours before their publication, so it is virtually impossible for the FOMC to react to financial market developments that occur during the narrow event window (Bauer & Swanson, 2023b, p. 111). A further requirement for the high-frequency approach is rational expectations in financial markets, so that asset prices incorporate new information shortly after its release (Swanson, 2011, pp. 7–8). Gürkaynak et al. (2005, pp. 80–82) find that asset prices incorporate most information from FOMC announcements within a 30-minute window that starts 10 minutes before and ends 20 minutes after the announcements.

The high-frequency event-study design is particularly suitable in the present context, because the focus is on immediate responses of the Treasury yield curve rather than persistent macroeconomic effects. Hence, the loss of statistical power induced by the small magnitude of measured MPS in the high-frequency event window will not pose any major problems (Nakamura & Steinsson, 2018, p. 1284).

[1] For some parts of this paper, the distinction between conventional and orthogonal MPS will be important. To support readability, unless explicitly stated otherwise, the term MPS is used without this distinction and more generally refers to monetary policy surprises per se in the following.

2.2 Transmission to the Stock Market

This section looks at the transmission of MPS during FOMC announcements to the stock market. It starts with the predictions of standard monetary policy transmission mechanisms and then discusses relevant literature on CBI effects.

According to standard stock valuation methods, the fundamental value of a stock can be represented by the sum of the present value of expected future dividends it generates and its expected future sale price (Damodaran, 2011, pp. 35–58). Thus, the fair price of a stock at time t can be expressed as

$$P_t = \frac{D_{t+1}}{1 + r_{t+1}} + \frac{D_{t+2}}{(1 + r_{t+1})(1 + r_{t+2})} + \ ... \ + \frac{D_{t+n} + P_{t+n}}{(1 + r_{t+1})(1 + r_{t+2})...(1 + r_{t+n})}, \qquad (2)$$

where, for the first year, D_{t+1} dividends are expected and r_{t+1} is the discount rate of equity. The stock is sold at time $t + n$.

In this simplified framework, monetary policy can affect stock prices via two components of Equation (2). First, MPS are transmitted to interest rates and thus also to discount rates of equity (Bernanke & Kuttner, 2005; Mishkin, 1996, pp. 2–4). While a contractionary surprise should raise the discount rate of equity, an expansionary surprise is expected to reduce it. Second, monetary policy influences the economic outlook by changing borrowing costs and setting incentives for or against consumption and investment (Jarociński & Karadi, 2020, p. 2; Mishkin, 1996, pp. 6–8). Under the standard transmission mechanism, a contractionary surprise worsens the economic outlook, which decreases expected future dividends. In contrast, an expansionary surprise should improve the economic outlook and increase expected future cash flows. Importantly, both effects work in the same direction under the standard monetary policy transmission mechanism. Contractionary MPS increase discount rates and reduce expected cash flows, both of which lower stock prices. Expansionary MPS do the opposite. Accordingly, the standard transmission mechanism unambiguously predicts negative co-movement between MPS and stock prices. However, empirically, stock prices often move in the same direction as MPS in high-frequency event windows around FOMC announcements. Jarociński and Karadi (2020, p. 2) find positive co-movement for about a third of FOMC announcements during the last three decades.

Jarociński and Karadi (2020) explain this positive co-movement between MPS and stock prices with the existence of CBI effects. Seminal work on the topic shows that Fed inflation forecasts outperform commercial inflation forecasts, indicating that there is an informational advantage of the central bank over the market (Romer & Romer, 2000). Further work discusses to what extent this informational advantage influences the transmission of MPS to economic and financial variables (Jarociński & Karadi, 2020; Nakamura & Steinsson, 2018). These authors found that

the superior information of central banks may be the reason for the effects of MPS that appear puzzling from the point of view of standard monetary policy transmission models. In particular, if market participants know that central banks possess superior information on economic fundamentals, they will try to derive information from the central bank's actions. As central banks pursue a stable price level, they aim to stabilize the business cycle and often counteract current economic developments. Therefore, a surprise monetary tightening or contemporaneously released communication can signal to the market that the economy is weak, whereas a surprise monetary easing may indicate the opposite. Returning to Equation (2), this means that the effect of monetary policy on the economic outlook and expected dividends is reversed, making the response of the stock price ambiguous (Bauer & Swanson, 2023a, p. 687; Jarociński & Karadi, 2020). As indicated, information about economic fundamentals may not only be revealed by the policy decision itself, since central banks have a variety of communication tools. For example, Swanson and Jayawickrema (2024) mention FOMC announcements, meeting minutes releases, speeches, press conferences, and congressional testimony by the Fed Chair. The present focus on 30-minute windows around FOMC announcements implies that only responses to the FOMC announcement itself are captured. Within FOMC announcements, information may be disclosed to the market by the policy decision itself, as it is based on the information available to the central bank. In addition, the 30-minute windows capture effects of the contemporaneously released FOMC statement and the likewise contemporaneously, but only quarterly, released Summary of Economic Projections (SEP). Consequently, CBI effects can explain positive co-movement between MPS and stock prices in the current high-frequency setting, if the effects of superior information within the policy decision itself, the FOMC statement, or the SEP dominate the effect on the discount rate of equity (Bauer & Swanson, 2023a, p. 687).

After Romer and Romer (2000), a considerable number of researchers tried to find out whether central banks indeed have information about economic fundamentals that is superior to information on the market. Recent contributions suggest that informational advantages do not exist systematically, but instead depend on the economic variable that is forecast, economic conditions, and the time period considered (Binder & Sekkel, 2023). Furthermore, Bauer and Swanson (2023a) provide empirical and survey evidence that speaks against informational advantages. These recent findings challenge the explanation of positive co-movement between MPS and stock prices with CBI effects. Instead, Bauer and Swanson (2023a) argue that the Fed and market participants respond to the same information, which is publicly available before the FOMC announcement. According to their view, MPS arise because the market systematically underestimates the Fed's response to news, such that the central bank does not reveal new information about economic fundamentals, but rather about its own reaction function to economic news (Bauer & Swanson, 2023a). However, the critique does not deliver a clear alternative explanation for observed positive co-movement between MPS and stock prices in the high-frequency event window. Rational expectations imply that asset prices incorporate all publicly

available information immediately (Swanson, 2011, p. 7). The information available prior to the FOMC announcement should therefore be incorporated into asset prices before the start of the event window, which means that it should not produce positive co-movement between MPS and stock prices. Furthermore, the critique must be understood in the context of a broader debate. Central banks can influence the expectations of commercial forecasters and market participants, even if they do not have better information on economic fundamentals (Binder & Sekkel, 2023). Hubert (2015) finds that central bank forecasts have an influence on private forecasts, which is unlikely to be attributable to a difference in forecast performance. He argues that central bank forecasts may also contain information that is not superior but different from the information on the market.

As a consequence, it is unclear whether the positive co-movement between MPS and stock prices can be explained with CBI effects. The aim of this paper is to find evidence that is consistent with the existence of CBI effects on FOMC announcements to contribute to the debate. Based on the literature reviewed in this section and following the approach of Jarociński and Karadi (2020, p. 15), FOMC announcements with positive co-movement are classified as information-dominated events, while FOMC announcements that exhibit negative co-movement are classified as policy-dominated events. The strategy relies on the assumption that, if positive co-movement between MPS and stock prices reflects CBI effects, then the Treasury yield curve responses on these FOMC announcements should also display patterns that are consistent with the information-related transmission of MPS. Thus, the classification itself is not interpreted as evidence of CBI effects but is used to compare the reaction of the Treasury yield curve to MPS between event types.

2.3 Transmission to the U.S. Treasury Market

U.S. Treasuries are issued by the U.S. Department of the Treasury to finance government spending. Consequently, Treasury yields represent the borrowing costs of the U.S. government. Due to their high liquidity and safety attributes, Treasuries play a central role in financial markets and are commonly used as benchmarks for the pricing of other assets (Krishnamurthy & Vissing-Jorgensen, 2012). Treasury yields at different maturities vary and therefore reflect the perception of financial markets of short- and long-term economic conditions (Gürkaynak & Wright, 2012). Thus, the Treasury yield curve responds to monetary policy and is also used as an indicator by monetary policymakers (Gürkaynak & Wright, 2012). Crucially, Treasuries with different maturities are affected by monetary policy through partly different channels, which makes the Treasury yield curve informative about the market interpretation of monetary policy (Gürkaynak et al., 2005; Hansen et al., 2019).

To understand the effects of these channels, one needs to understand some theory about the composition of yields with different maturities. A natural benchmark to explain the term struc-

ture of interest rates is the expectations hypothesis. It states that the yields of bonds with various periods until maturity are equal to the expected average short-term yield over the life of the long-term bond (Gürkaynak & Wright, 2012). Thus, Treasuries with shorter maturities are strongly influenced by current short-term yields, as they will soon reach maturity and are not affected by short-term yields of further periods. Meanwhile, long maturities are more affected by the expected path of short-term yields, since long-term Treasury yields incorporate expectations over a longer horizon. However, empirical evidence suggests that the expectations hypothesis cannot fully explain the yield curve (Gürkaynak & Wright, 2012). It cannot explain part of the movement in the longest maturities, as these yields reflect not only expected future short-term yields, but also the additional compensation required by investors to hold a bond with longer maturity (Hanson & Stein, 2015). When including this term premium into the framework, Treasury yields can be represented as

$$y_t(n) = E_t\left(\frac{1}{n}\sum_{i=0}^{n-1} y_{t+i}^{(\text{short})}\right) + TP_t(n)\ , \tag{3}$$

where $y_t(n)$ represents the yield of a Treasury with n years to maturity at time t and the expected average of short-term yields until maturity is denoted by $E_t(\frac{1}{n}\sum_{i=0}^{n-1} y_{t+i}^{(\text{short})})$ (Gürkaynak & Wright, 2012, p. 339). Meanwhile, the term premium of the n-year Treasury at time t is represented by $TP_t(n)$ (Gürkaynak & Wright, 2012, p. 339). The first part of Equation (3) comes from the expectations hypothesis, while the second part adjusts for the empirically observed failure of the pure expectations hypothesis. Note that the second part may arise from uncertainty about risks such as inflation and economic downturns, market segmentation, or other investor preferences (Gürkaynak & Wright, 2012; Hanson & Stein, 2015).

It is hence of interest to examine how the importance of the components of Equation (3) changes between maturities, as this will have implications on how MPS are transmitted to Treasury yields at different maturities. Monetary policy affects Treasury yields across the maturity spectrum (Gürkaynak et al., 2005). However, because the components of Equation (3) are not equally important across the maturity spectrum, monetary policy may not affect all maturities uniformly. Even more so, different monetary policy instruments provoke different responses on the maturity spectrum (Hansen et al., 2019). MPS triggered by policy decisions and MPS triggered by communication tools are therefore not equally transmitted across Treasuries with different maturities. The following discussion draws on work by Kaminska et al. (2021), who estimate an affine term structure model and decompose MPS into three components: surprises in monetary policy action, surprises in the expected path of monetary policy, and surprises in uncertainty.

The action component is driven by surprise changes in the current policy rate (Kaminska et al., 2021, p. 57). As the maturity of Treasuries increases, the effects of this component decline (Kaminska et al., 2021, p. 58). Consequently, short-term Treasury yields are more strongly

influenced by the action component than longer-term Treasury yields (Gürkaynak et al., 2005, pp. 83–84; Kaminska et al., 2021, p. 58). This is consistent with the expectations hypothesis, as short-term Treasury yields reflect expectations over a short horizon, which makes them more closely related to the current policy rate. As its name implies, the expected path component captures the effect of a surprise in the expected monetary policy path (Kaminska et al., 2021, p. 57). The component is driven not only by the current policy decision, but also by central bank communication and has stronger effects on long maturities than on short maturities (Gürkaynak et al., 2005; Kaminska et al., 2021, p. 58). This pattern can again be explained by the expectations hypothesis, as more yields enter the first part of Equation (3) for longer maturities. However, the influence of the expected path component on Treasury yields does not increase constantly with maturity. Instead, it is hump-shaped and peaks at intermediate maturities (Kaminska et al., 2021, p. 58). Lastly, the uncertainty component captures the effect of a surprise in uncertainty about the expected monetary policy path (Kaminska et al., 2021, p. 57). Such surprises may arise from news about inflation, economic risks, or other uncertainties and are therefore mainly driven by central bank communication (Hansen et al., 2019; Swanson, 2006). The effects of the uncertainty component on Treasuries increase with maturity, which can be explained by the fact that uncertainties are more relevant and unpredictable over longer time horizons (Hansen et al., 2019; Kaminska et al., 2021, p. 58). This means that the importance of the second part in Equation (3), which denotes the term premium, increases with maturity.

2.4 Implications

The previous discussion shows that Treasury yields are affected by surprises in the current policy rate, expected future policy, and uncertainty. It also shows that these components of MPS cause unequal patterns across the maturity spectrum of Treasuries. Such differing patterns imply that FOMC announcements that involve MPS with different compositions of action, expected path, and uncertainty components should provoke distinct responses on the Treasury yield curve. As a result, implications for the empirical analysis can be derived.

If FOMC announcements with positive co-movement between MPS and stock prices (classified as information-dominated) indeed reflect CBI effects, the policy decision, the FOMC statement, or the SEP should contain information that is superior to market information. This information could, for example, concern uncertainty or underlying economic conditions. Thus, MPS during information-dominated events are expected to exhibit systematically different proportional compositions of action, expected path, and uncertainty components compared to MPS associated with policy-dominated events. In particular, MPS on information-dominated events are expected to display relatively higher contributions from the expected path and uncertainty components. As a consequence, Treasury yield curve responses during information-dominated events should exhibit signs of transmission of these differing compositions of MPS, that is, signs

of information-related transmission channels. Since the policy path and uncertainty components are transmitted more strongly to intermediate and long maturities, stronger movements are expected at the long end of the Treasury yield curve during information-dominated events. Conversely, policy-dominated events are expected to be more closely associated with the action component of MPS and should therefore generate a comparatively stronger response on the short end of the yield curve. Thus, a differential response of the slope factor of the Treasury yield curve in response to MPS is expected between event types. More precisely, this leads to the following implication: if MPS contain relatively stronger expected path and uncertainty components during FOMC announcements with positive co-movement between MPS and stock prices than during FOMC announcements with negative co-movement, the Treasury yield curve is expected to get relatively steeper in response to contractionary MPS during these events.

This discussion is motivated by the aim of examining whether positive co-movement between MPS and stock prices is the only indicator consistent with the presence of CBI effects and builds on the preceding discussion of monetary policy transmission to the Treasury yield curve. Note that the Treasury yield curve responses cannot provide conclusive evidence on the existence of CBI effects. However, this analysis offers additional evidence on the debate by indicating whether Treasury yield curve responses are in accordance with the transmission of information-related components of MPS during information-dominated FOMC announcements.

3 Empirical Strategy

In this chapter, the data and methods used in the empirical investigation are presented.

3.1 Data and Sample Construction

The data source for the entire empirical analysis is the Bauer and Swanson (2023b) data set, provided on the website of the Center for Monetary Research of the Federal Reserve Bank of San Francisco (2025).

This data set covers FOMC announcements that span from February 1988 until December 2023, including a total number of 361 FOMC announcements. To reduce the noise in the data, the present analysis does not rely on the entire sample.[2] In particular, FOMC decisions were not explicitly announced prior to January 1994. Furthermore, according to the Board of Governors of the Federal Reserve System (2022), the FOMC statement was issued for every policy change starting in February 1995 and for every FOMC decision, regardless of whether there was a policy change or not, starting in January 2000. Because the FOMC statement and its potential to carry information are crucial for the present analysis, the main results are reported for a sample that starts in January 2000 and ends in December 2023. Within this sample period, 12 FOMC announcements between March 2020 and December 2020 are excluded from the analysis because the authors did not report orthogonal MPS. The reason for this is likely to be considerable noise in macroeconomic information as a consequence of the Covid-19 pandemic (Bauer & Swanson, 2023b, p. 119). In addition, the announcements on 17 September 2001 and 14 December 2010 are excluded. The first was the initial day of trading after the terrorist attacks, and in the latter, the stock response is missing. In the remaining sample of 192 FOMC announcements, all variables required for the main analysis are available. These variables are orthogonal MPS computed by the authors, raw data on high-frequency changes of the E-mini S&P 500, and raw data on changes in Treasury yields with maturities of 2, 5, 10, and 30 years.[3] For additional sensitivity checks, the sample period will be extended. These checks will also require conventional MPS.

As discussed in Section 2.1, the present analysis is based on high-frequency identification. Fol-

[2] For reproducibility, the entire code used in the empirical analysis is available in a public GitHub repository accessible over the following link: https://github.com/ericoehler/bachelor-thesis-code.git.

[3] Given the limited availability of high-frequency data and the emphasis on contemporaneous responses, the analysis is restricted to the Treasury data provided by Bauer and Swanson (2023b). Ideally, the term structure of interest rates would be represented by zero-coupon yields, which, however, must be estimated for specific maturities (Gürkaynak et al., 2006, pp. 2–3).

lowing the prominent approach of Gürkaynak et al. (2005), this paper focuses on 30-minute event windows that start 10 minutes before and end 20 minutes after FOMC announcements. For every FOMC announcement, all asset price responses used or analyzed in this paper are measured in the same 30-minute event window, which ensures that all variables reflect the same information arrival. Therefore, the relevant assumption is that changes in Treasury yields, stock prices, and interest rate futures within the 30-minute event windows are driven by the respective FOMC announcements.

3.2 Measurement of Monetary Policy Surprises

The present analysis relies on MPS constructed by Bauer and Swanson (2023b). As they are the main explanatory variable in the event-study regression, this section reviews their construction. Importantly, the authors provide two measures: conventional MPS and orthogonal MPS. Both measures of MPS are computed on the basis of high-frequency changes in Eurodollar futures in the 30-minute event windows around FOMC announcements. As discussed in Section 2.1, this makes it possible to isolate the unanticipated part of monetary policy, which is necessary to identify the real effects of monetary policy on asset prices (Bauer & Swanson, 2023b, p. 111; Kuttner, 2001). Before their discontinuation in April 2023, Eurodollar futures were priced based on the U.S. Dollar London Interbank Offered Rate (Brennan et al., 2024, p. 5). This interest rate was offered on the money market to exchange short-term dollar liquidity between financial intermediaries. Eurodollar futures are therefore closely related to expectations about short-term dollar interest rates and suitable for measuring MPS from FOMC announcements.

Bauer and Swanson (2023b, p. 103) perform three steps to construct conventional MPS. First, the high-frequency changes in implied rates of four Eurodollar futures contracts, covering the current quarter and the next three quarters ahead of the FOMC announcement, are measured over the event window.[4] Second, these high-frequency changes are used as input data for principal component analysis (PCA). Then, the first principal component is rescaled such that a one-unit change corresponds to a one-percentage-point change in the contract with the longest horizon. By construction, positive MPS are contractionary surprises and negative MPS are expansionary surprises. Moreover, the authors argue that the first principal component comprises movements in all four contracts and can be interpreted as a common surprise factor on interest rate futures up to one year ahead (Bauer & Swanson, 2023b, p. 103). As required for this analysis, they mention that these features make the first principal component suitable for capturing different components of MPS (Bauer & Swanson, 2023b, p. 103). Furthermore, PCA reduces the dimensionality from four correlated futures responses to one single measure of MPS.

[4]Because of the discontinuation of Eurodollar futures, the authors replace them with Secured Overnight Financing Rate Futures starting from January 2023 onward. The underlying rate of these futures is the successor of the U.S. Dollar London Interbank Offered Rate.

However, Bauer and Swanson (2023b) observe that conventional MPS are predictable with public macroeconomic information available prior to FOMC announcements. If both MPS and the responses of observed financial variables are correlated with this public information, concerns about endogeneity arise that may also cause illusory CBI effects (Bauer & Swanson, 2023b, pp. 121, 150). Thus, they provide an alternative measure by removing the component of conventional MPS that is predictable and calling it orthogonal MPS. In particular, they regress conventional MPS on the relevant macroeconomic information and take the residuals as the new measures of MPS. Orthogonal MPS can therefore be represented as

$$mps_t^{\perp} = mps_t^{conventional} - \hat{\alpha} - \hat{\beta}' \cdot \mathbf{V}_{t-} \,, \tag{4}$$

where $\mathbf{V}_{t-}$ represents the vector of public information before the announcement and $\hat{\beta}'$ denotes the vector with the estimated coefficients (Bauer & Swanson, 2023b, p. 113). Thus, these new measures of MPS are orthogonal to the relevant public information, which mitigates endogeneity concerns. In the present empirical analysis, orthogonal MPS are used, as they are the cleaner measure of MPS. Nevertheless, conventional MPS are applied as a robustness check because they remain appropriate in high-frequency event-study designs that estimate the effects of monetary policy on financial markets (Bauer & Swanson, 2023b, pp. 110–112). Still, if not explicitly stated otherwise, the term MPS is used without differentiation between the two measures and refers to surprises in monetary policy in general.

3.3 Event Classification

The responses of the stock market are also measured in the 30-minute event windows and reported by Bauer and Swanson (2023b). They report stock responses based on changes in the E-mini S&P 500 futures contract.[5] The authors approximate the percentage change in the E-mini S&P 500 as

$$\Delta SP500_t = 100 \cdot (ln(P_{t+20}) - ln(P_{t-10})) \,, \tag{5}$$

where P_{t-10} and P_{t+20} represent the futures prices ten minutes before and 20 minutes after the FOMC announcement t (Bauer & Swanson, 2023b, p. 149; Swanson & Jayawickrema, 2024, p. 13). Thus, a positive value in $\Delta SP500_t$ corresponds to an increase in the futures price, which is an increase in stock prices.

As discussed in Section 2.4, the objective is to compare the Treasury yield curve responses to MPS between FOMC announcements with positive and negative co-movement of MPS and

[5]The E-mini contract provides higher liquidity and longer trading hours than the original S&P 500 futures contract (Swanson & Jayawickrema, 2024, p. 13).

stock prices. Positive co-movement means that interest rates, as measured by MPS, and stock prices move in the same direction. Negative co-movement refers to the opposite case, where interest rates and stock prices move in reverse directions. FOMC announcements with negative co-movement between MPS and stock prices are classified as policy-dominated events, and announcements with positive co-movement are classified as information-dominated events. Subsequently, an indicator variable for the regression is constructed such that

$$D_t^{info} = \begin{cases} 1 & \text{if} \quad mps_t^{\perp} \cdot \Delta SP500_t > 0 \\ 0 & \text{if} \quad mps_t^{\perp} \cdot \Delta SP500_t < 0 \\ NA & \text{if} \quad mps_t^{\perp} \cdot \Delta SP500_t = 0 \,, \end{cases} \tag{6}$$

where $D_t^{info} = 1$ denotes an information-dominated event and $D_t^{info} = 0$ classifies a policy-dominated event. This classification strategy is motivated by Jarociński and Karadi (2020, p. 15) and their poor man's sign restriction. The advantage of this approach is that it can be directly applied in a high-frequency event-study setting, which is crucial for the current analysis of immediate Treasury yield curve responses around FOMC announcements. Moreover, Jarociński and Karadi (2020, pp. 15–16) show that this simple sign-based classification strategy and their full sign-restriction identification produce similar impulse responses in their structural vector autoregression specification. Nevertheless, it must be recognized that this classification strategy identifies only the dominant mechanism operating during a given event, rather than isolating pure information shocks or pure policy shocks (Jarociński & Karadi, 2020, pp. 15–17). As a consequence, information-dominated events likely still contain standard transmission effects, and policy-dominated events likely still contain information effects. This imperfect separation may attenuate the estimated differences between event types and is therefore an important limitation of the empirical strategy.

Furthermore, it is also important to note that the classification is not interpreted as evidence for CBI effects; rather, it provides the basis for the subsequent analysis, with the aim of finding further evidence on this exact debate. Its main purpose is to differentiate between FOMC announcements for the analysis of Treasury yield curve responses. More precisely, if positive co-movement between MPS and stock prices reflects CBI effects, the classification allows the analysis to compare events in which these effects are more important to events where they are less important.

3.4 Treasury Yield Curve Factor Construction

Raw data on changes in yields of 2-year, 5-year, 10-year, and 30-year Treasuries during the 30-minute event windows are measured in percentage points and again provided by Bauer and

Swanson (2023b). However, the current objective is to capture systematic patterns in Treasury yield changes across maturities and to compare these responses between FOMC announcements. Thus, analyzing these correlated yield changes separately for each maturity is less informative, and the application of PCA is more promising.

PCA allows the number of variables in the data to be reduced, while retaining most of the original information (Patel et al., 2018, p. 5). For changes in Treasury yields, the literature observes that the loss of information induced by PCA is small, since the first three principal components account for more than 90 % of the variation in the data (Litterman & Scheinkman, 1991, p. 57; Piazzesi, 2010, p. 737). This is achieved by summarizing the correlated original variables into uncorrelated principal components that are linear combinations of the original variables and capture the dominant patterns of variation in the data (Patel et al., 2018, pp. 4–5). The first principal component is the linear combination of variables that explains the maximum possible variance in the data (Patel et al., 2018, p. 5). Each subsequent principal component explains the largest possible share of the remaining variance, subject to the condition of being uncorrelated with the previous principal components (Patel et al., 2018, p. 5). Therefore, PCA orders the extracted components by explanatory power. In the present approach, PCA is applied to the centered matrix $X_c \in \mathbb{R}^{n\times 4}$, where the rows represent n FOMC announcements,[6] and the columns represent the four maturities. Note that because all Treasury yield changes are measured in the same unit, PCA is performed on the centered but not standardized data matrix (Jolliffe & Cadima, 2016, pp. 5–6).

In statistical software, PCA is usually computed via the singular value decomposition of the centered data matrix X_c. Following Jolliffe and Cadima (2016, p. 3), with the singular value decomposition one can decompose the centered data matrix such that

$$X_c = U\Sigma V^\top , \tag{7}$$

where $U \in \mathbb{R}^{n\times r}$ and $V \in \mathbb{R}^{m\times r}$ are matrices with orthonormal columns, $\Sigma \in \mathbb{R}^{r\times r}$ is a diagonal matrix containing the singular values, and $r = \operatorname{rank}(X_c) \leq 4$. Singular values are ordered by decreasing magnitude, which determines the order of the corresponding singular vectors. The columns of V are the right singular vectors and contain the principal component loadings (Jolliffe & Cadima, 2016, p. 3). Jolliffe and Cadima (2016, p. 3) project the centered data matrix onto these loadings to obtain the principal components

$$Z = X_c V . \tag{8}$$

[6] In the main sample, the number of FOMC announcements is 192. For the sensitivity analyses this number varies.

Crucially, the loadings determine how strongly each original variable contributes to a given principal component and therefore help to identify its economic interpretation (Piazzesi, 2010, p. 737; Jolliffe & Cadima, 2016, pp. 2–3).

In the present analysis, PCA is applied to high-frequency Treasury yield changes. Hence, the loadings on a principal component define how changes in different Treasury yields affect the respective principal component. Empirical and theoretical research supports the interpretation of the first three principal components of Treasury yields as level, slope, and curvature factors of the yield curve (Litterman & Scheinkman, 1991, pp. 57–58; Lord & Pelsser, 2007; Piazzesi, 2010, pp. 737–738). Since the present investigation is based on changes in Treasury yields, the principal components extracted should be interpreted as changes in the underlying factors of the Treasury yield curve, rather than as the levels of these factors themselves (Piazzesi, 2010, pp. 736–737). This interpretation is motivated by the structure of the factor loadings associated with the principal components (Litterman & Scheinkman, 1991, pp. 57–58; Lord & Pelsser, 2007; Piazzesi, 2010, pp. 737–738).

According to the literature cited above, the first principal component is interpreted as the level factor because its loadings are relatively flat across maturities. Therefore, it corresponds to parallel shifts in the yield curve. The second principal component is understood as the slope factor because its loadings decrease or increase with maturity, so it reflects changes in the steepness of the yield curve. The third principal component is called the curvature factor because its loadings are hump-shaped across maturities. With the hump at intermediate maturities, this means that it captures how intermediate maturities respond compared to the short and long ends of the yield curve. Importantly, the signs of principal components and their loadings are arbitrary and can be reversed without changing the statistical content (Jolliffe & Cadima, 2016, p. 3).[7] For interpretation purposes, this paper follows the sign convention in Piazzesi (2010, p. 737): loadings on the level factor are positive, loadings on the slope factor are initially positive and turn negative with increasing maturity, and loadings on the curvature factor are positive for intermediate maturities but negative at the short and long end. In Section 4.1, a figure reports the loadings of the four maturities on the first three principal components obtained from the current sample.

In this way, the extraction of the first three principal components of high-frequency Treasury yield changes facilitates the comparison of responses of identified yield curve factors between information-dominated and policy-dominated events. Furthermore, unlike parametric approaches to model the yield curve, such as the Nelson-Siegel model by Nelson and Siegel (1987), PCA is an empirical approach and does not require an imposed structure. These characteristics motivate the application of PCA in the present analysis.

[7] If the signs are reversed (multiplied by -1), the principal component and its loadings must both be reversed.

3.5 Regression Specification

To explore the relationship between MPS and changes in the first three principal components of Treasury yield changes, interpreted as responses of Treasury yield curve factors, the responses of these factors are separately regressed on orthogonal MPS. The baseline specification is based on the high-frequency literature as discussed in Section 2.1. Thus, the following event-study regressions are estimated:

$$PC_{t,k} = \beta_{0,k} + \beta_{1,k} \cdot mps_t^{\perp} + u_{t,k} , \tag{9}$$

where $PC_{t,k}$ denotes the score of the principal component k of the high-frequency Treasury yield changes around FOMC announcement t, and $mps_t^{\perp}$ represents the orthogonal MPS corresponding to this announcement. Consequently, the coefficient $\beta_{1,k}$ captures the sensitivity of changes in the Treasury yield curve factor k to orthogonal MPS. Estimating this specification separately for the three principal components provides a benchmark assessment of how the level, slope, and curvature factors respond to orthogonal MPS. These responses can then be compared with existing literature to provide support for the currently adopted empirical strategy.

In order to differentiate between information-dominated and policy-dominated events, the regression specification is subsequently extended by interacting with the classification indicator D_t^{info} from Equation (6). The extended specification can be written as

$$PC_{t,k} = \beta_{0,k} + \beta_{1,k} \cdot mps_t^{\perp} + \alpha_k \cdot D_t^{info} + \gamma_k \cdot (D_t^{info} \cdot mps_t^{\perp}) + \epsilon_{t,k} , \tag{10}$$

with $D_t^{info} = 1$ for information-dominated events and $D_t^{info} = 0$ for policy-dominated events. Again, the estimated coefficients capture how orthogonal MPS affect the shape of the Treasury yield curve, as summarized by the principal components. In the following, the interpretation of the coefficients in this extended specification relies on the explanations in Wooldridge (2020, pp. 232–234). $\beta_{1,k}$ measures the response of changes in the yield curve factor k to orthogonal MPS during policy-dominated events. The coefficient α_k captures differences in the intercept between information-dominated and policy-dominated events for changes in the Treasury yield curve factor k. Thus, it is the difference in the change in the Treasury yield curve factor k between event types, if $mps_t^{\perp} = 0$. Meanwhile, the interaction coefficient γ_k captures the differential response of changes in the Treasury yield curve factor k to orthogonal MPS during information-dominated events compared to policy-dominated events. Therefore, the sensitivity of changes in the Treasury yield curve factor k to orthogonal MPS during information-dominated events is given by $\beta_{1,k} + \gamma_k$. Hence, if γ_k is statistically significantly different from zero, this provides direct evidence for a differential response of the Treasury yield curve factor k to orthogonal

MPS between the two event types. Due to the direct significance test, this research design is preferred to performing separate regressions for information-dominated and policy-dominated events. Due to the implications developed in Section 2.4, the main coefficient of interest is γ_{slope}, as it captures the expected heterogeneous response across the maturity spectrum of Treasuries during information-dominated events.

For interpretation purposes, it is important to remember the sign convention chosen in Section 3.4. Positive coefficients on the three Treasury yield curve factors imply that contractionary MPS are associated with an upward shift, a flatter slope, and a greater curvature of the yield curve. Furthermore, the interpretation of the regression coefficients needs careful consideration, as the Treasury yield curve factors are new variables composed of linear combinations of the original variables. Because in the present case PCA is applied to the centered but not standardized data matrix and the changes in the four Treasuries are all measured in percentage points, the first three principal components retain this unit of the original variables (Jolliffe, 2002, p. 274). Consequently, the estimated coefficients represent the impact on the respective principal components in percentage points (or basis points).

4 Results

In this chapter, the empirical findings are presented. First, descriptive evidence, such as summary statistics on orthogonal MPS, event classification, and PCA of Treasury yield changes, is reported to support the most important steps of the investigation. Then, the regression results from the baseline and the extended specification are displayed, subjected to sensitivity checks, and interpreted in light of the present research question.

4.1 Descriptive Evidence

Descriptive evidence is reported for the filtered sample used in the main analysis, which covers FOMC announcements between January 2000 and December 2023. Further exclusions mentioned in Chapter 3.1 leave the filtered sample at 192 observations.

Table 1 reports descriptive statistics for orthogonal MPS and the event classification.

Table 1: Orthogonal MPS and Event Classification

Panel A: $mps^{\perp}$ (in basis points)		**Panel B: Event Classification**	
Mean	0.406	Information-dominated	67
SD	5.202		(34.9%)
Median	0.326	Policy-dominated	125
Min	-23.195		(65.1%)
Max	16.07		

Notes: The table reports descriptive statistics for orthogonal MPS and the event classification based on the main sample of FOMC announcements covering 192 observations. Orthogonal MPS are reported in basis points.

Orthogonal MPS serve as the main explanatory variable in the regression analysis. As discussed in Section 3.2, positive values are contractionary surprises and negative values indicate expansionary surprises. Because they are signed surprises, the mean and median are around zero and of limited interest here. More relevant is the extent of variation between different FOMC announcements. The standard deviation is 5.202 basis points and the sample ranges from -23.195 to 16.07 basis points, which indicates that orthogonal MPS are typically small but still display meaningful variation across events. The amount of variation observed here is in agreement with the variation reported by Nakamura and Steinsson (2018, p. 1284) for their sample. Such variation suggests that orthogonal MPS are suitable for the analysis of contemporaneous financial market responses to FOMC announcements (Nakamura & Steinsson, 2018, p. 1284). Thus,

this variation supports the present assessment of Treasury yield curve responses to monetary policy. Moreover, the fact that the measured MPS are always closer to zero than the usual 25-basis point policy step illustrates the importance of focusing on the unanticipated component of monetary policy. Otherwise, the effects of monetary policy would likely be estimated with bias (Kuttner, 2001). Meanwhile, event classification is crucial to distinguish between information-dominated and policy-dominated events, that is, events with positive and negative co-movement between MPS and stock prices. For this reason, the frequencies of both event types are shown. Approximately 35 % of the FOMC announcements in the sample are classified as information-dominated, while the remaining, roughly 65 %, are policy-dominated. The share of information-dominated events in the present sample is very similar to the fraction of events with positive co-movement between short-term interest rate surprises and stock prices reported by Jarociński and Karadi (2020, pp. 7–8) for their sample. Furthermore, although information-dominated events are less frequent, both event types remain sufficiently represented in the sample to allow a meaningful comparison of their Treasury yield curve responses.

Table 2 reports the share of the total variation in Treasury yield changes explained by each of the first three principal components, as well as their cumulative explanatory power.

Table 2: Explained Variance of Principal Components

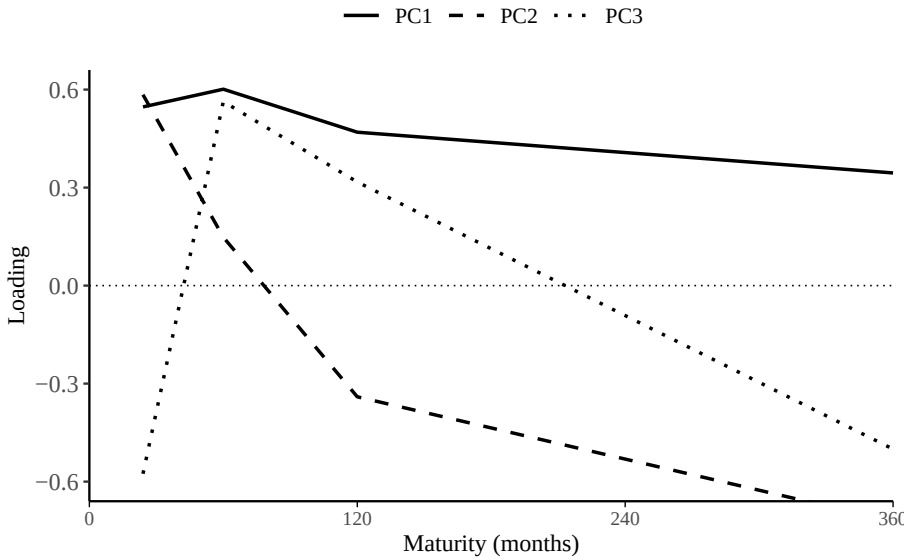
Component	Explained Variance (%)	Cumulative Variance (%)
PC1	86.07	86.07
PC2	11.96	98.02
PC3	1.73	99.76

Notes: The table reports the shares of total variance of high-frequency Treasury yield changes explained by their first three principal components.

As expected, the shares of explained variation decrease across components, with the first principal component accounting for the vast majority of total variation. Together, the first three principal components account for more than 99 % of the variation in observed Treasury yield changes, which means that loss of information is minimal. Compared with the evidence reported in Piazzesi (2010, p. 737), the first principal component explains a relatively large share in the present sample and the third principal component a relatively small one. A plausible explanation for this is the more limited number of Treasuries observed here, as the present PCA is based on changes in only four different maturities, while Piazzesi (2010, p. 736) relies on a set of 11 yields. Although the small share of variance explained by the third principal component does not make it irrelevant (Jolliffe, 1982), the interpretation of the regression results on the third principal component will remain cautious. The reason for this is that the representation of the curvature factor as the third principal component is less robust than that of the level and slope factors (Lord & Pelsser, 2007, pp. 20–23). This may be especially relevant because of the limited number of maturities currently available.

As discussed in Section 3.4, the loadings on the first three principal components justify their interpretation as the level, slope, and curvature factors of the Treasury yield curve (Litterman & Scheinkman, 1991, pp. 57–58; Lord & Pelsser, 2007; Piazzesi, 2010, pp. 737–738). Figure 1 plots the loadings of high-frequency Treasury yield changes in the present sample on the first three principal components.

Figure 1: Loadings on Principal Components across Maturities



Source: Own calculations based on Bauer and Swanson (2023b); presentation follows Piazzesi (2010, p. 738).

Again, the small set of maturities is the most likely reason that the plotted lines are less smooth than the ones visualized by Piazzesi (2010, p. 738). However, more crucial are the signs of the loadings on the three principal components, which motivate the interpretation as the level, slope, and curvature factors. The loadings on the first principal component are of the same sign and of similar magnitude. This validates its interpretation as the level factor of the Treasury yield curve. Loadings on the second principal component are of opposite sign at the short and long end, which motivates its interpretation as the slope factor. For the third principal component, the observed loadings are hump-shaped, implying that intermediate maturities move differently from short and long maturities. This is why it would theoretically be interpreted as the curvature factor; however, in the present analysis this interpretation remains cautious. Due to the sign convention outlined in Section 3.4, all maturities exhibit positive loadings on the first principal component. In contrast, on the second principal component, only the 2-year and 5-year Treasury yield changes load positively, whereas the 10-year and 30-year changes load negatively. Meanwhile, on the third principal component, the 5-year and 10-year Treasury yield changes have positive loadings, whereas the 2-year and 30-year changes load negatively.

4.2 Baseline Results

As a baseline, each of the three principal components is regressed separately on orthogonal MPS, as in Equation (9). Table 3 reports these regression results with the principal components labeled level, slope, and curvature for the main sample from January 2000 to December 2023.

Table 3: Baseline Specification

	Level	Slope	Curvature
$mps^{\perp}$	1.063***	0.257***	-0.044***
	(0.131)	(0.037)	(0.016)
Observations	192	192	192
R^2	0.428	0.180	0.037

Notes: The table reports regression results of the first three principal components of Treasury yield changes on orthogonal MPS. Thus, columns correspond to changes in the first, second, and third principal component. Robust standard errors (HC1) are reported in parentheses. Statistical significance is based on the corresponding robust p-values. *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

The estimated coefficients for the three factors are highly statistically significant, with p-values below 0.01. β_{level} implies that an increase of one basis point in orthogonal MPS is associated with an increase of 1.063 basis points in the first principal component score and therefore in the linear combination of Treasury yield changes that is classified as the response of the level factor. Under the chosen sign convention, this implies that contractionary MPS are associated with a higher overall level of Treasury yields, whereas expansionary MPS are associated with a lower level. This result is consistent with the literature, which finds that MPS move Treasury yields across maturities and in the same direction as the surprise (Gürkaynak et al., 2005; Kuttner, 2001).

β_{slope} indicates that an increase in orthogonal MPS of one basis point is associated with an increase of the second principal component by 0.257 basis points. Again, this means that the linear combination of Treasuries classified as the response of the slope factor increases by 0.257 basis points. Given the loading pattern on the second principal component, with positive loadings at short maturities and negative loadings at long maturities, an increase in the second principal component implies that short maturities rise more strongly relative to long maturities. Therefore, because β_{slope} is positive, contractionary MPS are associated with a flatter yield curve and expansionary MPS with a steeper yield curve. This stronger response of short maturities to MPS appears interesting in the context of literature that emphasizes that the short end of the yield curve is more closely related to the current policy rate compared to the long end (Hansen et al., 2019; Kaminska et al., 2021, p. 58). The question arises whether the response of the second principal component to MPS differs for information-dominated events. As hypothesized in Section 2.4, MPS could exhibit a different composition of action, expected path, and uncertainty components during these events.

For the curvature factor, the coefficient is negative, indicating that contractionary MPS are associated with a less pronounced hump at intermediate maturities. This result suggests that intermediate maturities, compared to the short and long ends of the yield curve, respond differently. However, as discussed in Section 4.1, it is unclear whether this result can be interpreted meaningfully.

In general, the baseline results show that the responses of the factors of the Treasury yield curve, measured by the first three principal components of high-frequency changes in Treasury yields, are sensitive to MPS. The findings support the empirical strategy, as the estimated responses of the level and slope factor are consistent with the related literature. At the same time, the baseline specification does not distinguish between information-dominated and policy-dominated events.

4.3 Main Results: Event Heterogeneity

With the aim of identifying potential differences between event types, the three principal components are separately regressed in the extended regression specification from Equation (10). In Table 4, the results are reported for the main sample from January 2000 to December 2023.

Table 4: Extended Specification

	Level	Slope	Curvature
$mps^{\perp}$	1.034***	0.282***	-0.041**
	(0.157)	(0.042)	(0.019)
D^{info}	0.007	-0.007*	-0.001
	(0.008)	(0.004)	(0.002)
$mps^{\perp} \times D^{info}$	0.155	-0.134**	-0.020
	(0.214)	(0.067)	(0.029)
Observations	192	192	192
R^2	0.431	0.199	0.039

Notes: The table reports regression results of the first three principal components of Treasury yield changes on orthogonal MPS, an indicator for information-dominated events, and their interaction term. Thus, columns correspond to changes in the first, second, and third principal component. Robust standard errors (HC1) are reported in parentheses. Statistical significance is based on corresponding robust p-values. *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

The baseline coefficients, which are the estimated coefficients on orthogonal MPS, remain statistically significant and similar in magnitude. Since policy-dominated events form the reference category, these estimates indicate that, on policy-dominated events, contractionary MPS are still associated with a higher level and a flatter slope of the Treasury yield curve. For the level and curvature factors, neither the classification indicator nor the interaction term is statistically significant, suggesting that no statistically significant differences between information- and policy-dominated events are found for these factors.

In contrast, the main result is the statistically significant negative interaction coefficient for the slope factor. γ_{slope} assesses the differential sensitivity of the slope factor to orthogonal MPS during information-dominated events and has a p-value below 0.05. The coefficient indicates that the sensitivity of the slope factor to orthogonal MPS is attenuated for information-dominated events. During information-dominated FOMC announcements, an increase of one basis point in orthogonal MPS is associated with an increase in the slope factor of $\beta_{1,slope} + \gamma_{slope}$, which is $0.282 - 0.134 = 0.148$. Consequently, the response of the slope factor, although attenuated, remains positive on information-dominated events. A robust Wald test of $H_0 : \beta_{1,slope} + \gamma_{slope} = 0$ rejects the null hypothesis at a p-value below 0.01. Therefore, the regression results indicate that on information-dominated events, contractionary MPS are still associated with a flatter Treasury yield curve; however, the flattening is less pronounced than during policy-dominated events. Similarly, expansionary MPS are associated with a steeper Treasury yield curve but with less pronounced steepening than during policy-dominated events. The coefficient of the classification indicator, α_{slope}, is weakly statistically significant. In the present analysis, this is less interesting as it indicates a lower intercept of responses of the slope factor during information-dominated events, which is not within the focus of this paper.

In summary, statistically significant differences between the two event types can be observed for the change in steepness of the yield curve in response to orthogonal MPS, but not for changes in the overall level of yields or the relative response of medium maturities compared to the ends.

4.4 Sensitivity Analysis

To review the robustness of the previous results and the validity of the empirical strategy, both are subjected to a sensitivity analysis. All the regression results from this sensitivity analysis are reported in the appendix.

In the first robustness check, statistical inference is modified and now relies on more conservative heteroskedasticity-consistent standard errors than previously. In particular, previously used heteroskedasticity-consistent standard errors of type 1 (HC1) are replaced by heteroskedasticity-consistent standard errors of type 3 (HC3). While HC1 is often set as standard in statistical software, the more conservative HC3 can be more suitable in small samples (Long & Ervin, 2000). The statistical significance of the previous results based on these more conservative standard errors is reported in Table A.1. Naturally, the estimates are not affected. In addition, the significance of orthogonal MPS on changes in the three Treasury yield curve factors remains unchanged with p-values below 0.01. Moreover, α_{slope} and γ_{slope} also remain statistically significant, but the p-value of the interaction coefficient increases to roughly 0.06. Thus, when using the more conservative HC3, the statistical significance of the differential sensitivity of the slope factor to MPS between event types is weaker but still present.

As a second robustness check, the extended regression is estimated for a longer sample period. Initially, the sample period starts in February 1995. As mentioned in Section 3.1, this is the time when FOMC statements started to be issued after policy rate changes, but not after policy decisions without policy rate changes. Hence, the sample spans from February 1995 to December 2023 and includes 233 FOMC announcements. Table A.2 reports these results. They look similar to the results of the first robustness check. Again, γ_{slope} is only weakly statistically significant, with a p-value close to 0.06. Here, the lower statistical significance of the interaction coefficient is likely attributable to increased noise, since, on top of the different communication regime, the E-mini S&P 500 contract was not available prior to September 1997 and has to be replaced with the original S&P 500 futures contract for observations before this date. This explanation appears particularly reasonable after looking at Table A.3, which reports the results estimated for the sample spanning from September 1997 to December 2023. Here, the p-value of the interaction coefficient is again below 0.05. In conclusion, the significance of γ_{slope} is somewhat robust to alternative sample periods and increased noise.

The next robustness check replaces orthogonal MPS with conventional MPS. Conventional MPS are used for the event classification and in the extended regression. According to Bauer and Swanson (2023b, pp. 110–112), conventional MPS are applicable for high-frequency event-study regressions. Regression results are reported in Table A.4. Again, the baseline results are validated, except for the curvature factor. Furthermore, γ_{slope} is of comparable size and weakly statistically significant with an increase in its p-value. Interestingly, when conventional MPS are used, the interaction coefficient for the level factor becomes highly statistically significant and increases strongly in size. As this is the only specification with a statistically significant interaction coefficient on the level factor, the result is likely to be attributable to the usage of conventional MPS. This suggests that the detected heterogeneity between event types is partly sensitive to the MPS measure, as it loads differently if the predictable component of MPS is not removed.

To test the validity of the empirical strategy, the analysis is now repeated with a stricter event classification. Because the classification border may seem arbitrary, the events that are closest to the classification border are excluded from the analysis. More precisely, the analysis excludes all FOMC announcements for which the absolute value of the stock response, the orthogonal MPS, or both are among the lowest 10 % of the sample. The idea for this comes from Jarociński and Karadi (2020, p. 7), who also exclude the events closest to the classification border. In Table A.5, these regression results are reported. They support the size and significance of the baseline coefficients. Again, the p-value of γ_{slope} increases, but the coefficient remains weakly statistically significant.

Finally, the validity of summarizing changes in the four Treasury yields by changes in principal components and interpreting them as Treasury yield curve factors is tested. Of particular interest

in the present analysis is the second principal component, which is the reason why it is replaced by more intuitive slope measures. To this end, the spread between long and short maturities is used. Table A.6 reports regression results for the responses of spreads between the 30-year and 2-year, the 30-year and 5-year, and the 10-year and 2-year Treasuries. Only the response of the widest spread, between the 30-year and the 2-year Treasury, shows weak statistical significance of the interaction term with a p-value below 0.1. However, while not statistically significant, the sign of the interaction coefficient is positive for all spread measures. This positive sign is in agreement with the results previously found with PCA. It implies that the spreads decrease less in response to contractionary MPS during information-dominated events than during policy-dominated events. Therefore, while on both event types short maturities increase more strongly in response to contractionary MPS, the difference between short and long maturities is relatively smaller on information-dominated events. Nevertheless, these results point out that the current representation of the Treasury yield curve may not be adequate. One plausible explanation is a lack of different maturities, as four Treasuries may not be sufficient to represent the Treasury yield curve.

In summary, the baseline estimates show strong robustness across specifications, except for the influence of MPS on the curvature factor, which underscores the importance of caution when interpreting the third principal component. Furthermore, all sensitivity analyses show some statistically significant differences between the two event types. The size of γ_{slope} is similar in all tests and its value is always negative, providing support for the attenuation of the response of the slope factor during information-dominated events. However, the p-values of the coefficients are of moderate size, and generally between 0.05 and 0.1. Repeating the analysis with conventional MPS shows that the Treasury yield curve factor on which differential effects between event types load may depend on the choice of the MPS measure. As orthogonal MPS are the cleaner measure, these surprises are generally preferred. In addition, the spread analysis only produces weakly statistically significant differences between event types for the widest spread measure in the present sample. Thus, the results on event heterogeneity also show some robustness to a variety of tests, but the evidence is less clear than it is for the baseline estimates.

4.5 Economic Interpretation

Treasury yield levels increase in response to contractionary MPS and decrease in response to expansionary MPS. Furthermore, the responses of the slope factor indicate that the reactions of long-term Treasuries are generally weaker on both event types. However, the central empirical result is the statistically significant negative interaction coefficient in the regression of the slope factor. While an increase in orthogonal MPS of one basis point increases the response of the slope factor by 0.282 basis points during policy-dominated FOMC announcements, this increase only amounts to 0.148 basis points during information-dominated events. Since under the cho-

sen sign convention an increase in the slope factor implies a decrease in steepness of the Treasury yield curve, the results indicate that long maturities exhibit relatively stronger responses to MPS during information-dominated events compared to their behavior during policy-dominated events. Thus, contractionary MPS are associated with a flattening of the Treasury yield curve, but the flattening is weaker on information-dominated FOMC announcements. The same logic applies to expansionary MPS, as they induce a similar pattern with movements in the opposite direction.

The implications from Section 2.4 predict a stronger response in long-term Treasury yields if MPS contain relatively stronger expected path and uncertainty components during FOMC announcements with positive co-movement between MPS and stock prices. During these announcements, the results indicate attenuated flattening in response to contractionary MPS and attenuated steepening in response to expansionary MPS. Hence, the results are consistent with a greater importance of information-related components, such as the expected path and uncertainty components, of MPS on information-dominated FOMC announcements.

The main objective of the present analysis is to find evidence on whether the response of the Treasury yield curve is consistent with the presence of CBI effects on information-dominated events. Expected path and uncertainty components of MPS may be caused by superior information released during FOMC announcements. Accordingly, the relatively stronger response of long maturities on information-dominated events is consistent with the CBI interpretation of positive co-movement between MPS and stock prices. However, the observed effects could also be due to information that is not superior but different from the information the market possesses (Hubert, 2015). In the present context, it is not possible to distinguish between the two types of information. Another argument is that the Fed does not release information about the economy, but rather about its own reaction function (Bauer & Swanson, 2023a). As stated by Kaminska et al. (2021, p. 58), surprises about the expected policy path also affect long maturities, but have their strongest influence on intermediate maturities. Thus, since the results on the third principal component cannot be adequately interpreted in the present analysis, this explanation is not ruled out either.

Overall, the results suggest that information-dominated events differ from policy-dominated events not only in their stock price response, but also in their Treasury yield curve response. However, the evidence has to be interpreted as consistent with CBI effects, rather than conclusive.

5 Conclusion

This paper contributes to the debate on CBI effects by assessing the responses of the U.S. Treasury yield curve to MPS. Specifically, it examines whether MPS during FOMC announcements with positive co-movement between MPS and stock prices also provoke distinct high-frequency responses of the Treasury yield curve.

The main finding is that the slope factor of the Treasury yield curve responds differently between FOMC announcements with positive and negative co-movement. Contractionary MPS are associated with a flattening of the Treasury yield curve on both event types. However, this flattening is attenuated if an FOMC announcement is accompanied by positive co-movement between MPS and stock prices, implying a relatively stronger response of Treasuries with long maturities. Although the p-value increases above 0.05 during most sensitivity analyses, the difference between event types remains weakly statistically significant and of the same sign. This pattern is consistent with different compositions of MPS between the two event types. In particular, it is consistent with relatively stronger contributions of the expected path and uncertainty components to MPS during FOMC announcements with positive co-movement.

Therefore, in addition to positive co-movement between MPS and the stock market, the response of the Treasury yield curve slope can be interpreted as consistent with CBI effects on information-dominated events. However, the evidence is not conclusive. More precisely, the observed pattern may not only be caused by the disclosure of superior information by the central bank, but also by the release of new information of equal quality or by revisions in the market's beliefs about the central bank's reaction function (Bauer & Swanson, 2023a; Hubert, 2015).

Furthermore, the analysis has two major limitations. First, the limited data availability in the high-frequency setting weakens the representation of the Treasury yield curve, as only four coupon-bearing Treasuries are available. Second, the sign-based classification strategy does not identify pure shocks of information or policy, which likely makes the results on the interaction coefficient less clear. Consequently, this paper provides a foundation for future research that investigates the response of the Treasury yield curve to MPS using more flexible data sources and a cleaner identification strategy to separate information and policy effects. Such research could focus on the differentiation between the effects of superior information, the effects of different information of equal quality, and the effects of information about the central bank's reaction function on the Treasury yield curve. Findings in this area could help monetary policymakers better assess the impact of their interventions.

References

Bauer, M. D., & Swanson, E. T. (2023a). An alternative explanation for the Fed information effect. *American Economic Review*, *113*(3), 664–700. https://doi.org/10.1257/aer.20201220

Bauer, M. D., & Swanson, E. T. (2023b). A reassessment of monetary policy surprises and high-frequency identification. *NBER Macroeconomics Annual*, *37*(1), 87–155. https://doi.org/10.1086/723574

Bernanke, B. S., & Kuttner, K. N. (2005). What explains the stock market's reaction to Federal Reserve policy? *The Journal of Finance*, *60*(3), 1221–1257. https://doi.org/10.1111/j.1540-6261.2005.00760.x

Binder, C. C., & Sekkel, R. (2023). *Central bank forecasting: A survey* (Staff Working Paper No. 2023-18). Bank of Canada. https://doi.org/10.34989/swp-2023-18

Board of Governors of the Federal Reserve System. (2022, January 28). *Federal Open Market Committee: Transcripts and other historical materials*. Retrieved April 27, 2026, from https://www.federalreserve.gov/monetarypolicy/fomc_historical.htm

Brennan, C. M., Jacobson, M. M., Matthes, C., & Walker, T. B. (2024). *Monetary policy shocks: Data or methods?* (Finance and Economics Discussion Series No. 2024-011). Board of Governors of the Federal Reserve System. https://doi.org/10.17016/FEDS.2024.011r1

Damodaran, A. (2011). *The little book of valuation: How to value a company, pick a stock and profit* (1st ed.). John Wiley & Sons.

Federal Reserve Bank of San Francisco. (2025, August 25). *Monetary policy surprises* [Data set]. Retrieved March 15, 2026, from https://www.frbsf.org/research-and-insights/data-and-indicators/monetary-policy-surprises/

Gürkaynak, R. S., Sack, B., & Swanson, E. T. (2005). Do actions speak louder than words? The response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, *1*(1), 55–93. https://www.ijcb.org/journal/v1n1/do-actions-speak-louder-words-response-asset-prices-monetary-policy-actions-and

Gürkaynak, R. S., Sack, B., & Wright, J. H. (2006). *The U.S. Treasury yield curve: 1961 to the present* (Finance and Economics Discussion Series No. 2006-28). Board of Governors of the Federal Reserve System. https://www.federalreserve.gov/pubs/feds/2006/200628/200628pap.pdf

Gürkaynak, R. S., & Wright, J. H. (2012). Macroeconomics and the term structure. *Journal of Economic Literature*, *50*(2), 331–367. https://doi.org/10.1257/jel.50.2.331

Hansen, S., McMahon, M., & Tong, M. (2019). The long-run information effect of central bank communication. *Journal of Monetary Economics*, *108*, 185–202. https://doi.org/10.1016/j.jmoneco.2019.09.002

Hanson, S. G., & Stein, J. C. (2015). Monetary policy and long-term real rates. *Journal of Financial Economics*, *115*(3), 429–448. https://doi.org/10.1016/j.jfineco.2014.11.001

Hubert, P. (2015). Do central bank forecasts influence private agents? Forecasting performance versus signals. *Journal of Money, Credit and Banking*, *47*(4), 771–789. https://www.jstor.org/stable/24500910

Jarociński, M., & Karadi, P. (2020). Deconstructing monetary policy surprises: The role of information shocks. *American Economic Journal: Macroeconomics*, *12*(2), 1–43. https://doi.org/10.1257/mac.20180090

Jolliffe, I. T. (1982). A note on the use of principal components in regression. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, *31*(3), 300–303. https://doi.org/10.2307/2348005

Jolliffe, I. T. (2002). *Principal component analysis* (2nd ed.). Springer. https://doi.org/10.1007/b98835

Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. *Phil. Trans. R. Soc. A*, *374*, Article 20150202. https://doi.org/10.1098/rsta.2015.0202

Kaminska, I., Mumtaz, H., & Šustek, R. (2021). Monetary policy surprises and their transmission through term premia and expected interest rates. *Journal of Monetary Economics*, *124*, 48–65. https://doi.org/10.1016/j.jmoneco.2021.07.009

Krishnamurthy, A., & Vissing-Jorgensen, A. (2012). The aggregate demand for Treasury debt. *Journal of Political Economy*, *120*(2), 233–267. https://doi.org/10.1086/666526

Kuttner, K. N. (2001). Monetary policy surprises and interest rates: Evidence from the Fed funds futures market. *Journal of Monetary Economics*, *47*(3), 523–544. https://doi.org/10.1016/S0304-3932(01)00055-1

Litterman, R. B., & Scheinkman, J. (1991). Common factors affecting bond returns. *The Journal of Fixed Income*, *1*(1), 54–61. https://math.nyu.edu/inmemoriam/avellaneda//Litterman1991.pdf

Long, J. S., & Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, *54*(3), 217–224. https://doi.org/10.1080/00031305.2000.10474549

Lord, R., & Pelsser, A. (2007). Level-slope-curvature - fact or artefact? *Applied Mathematical Finance*, *14*(2), 105–130. https://doi.org/10.1080/13504860600661111

Mishkin, F. S. (1996). *The channels of monetary transmission: Lessons for monetary policy* (NBER Working Paper No. 5464). National Bureau of Economic Research. https://doi.org/10.3386/w5464

Nakamura, E., & Steinsson, J. (2018). High-frequency identification of monetary non-neutrality: The information effect. *The Quarterly Journal of Economics*, *133*(3), 1283–1330. https://doi.org/10.1093/qje/qjy004

Nelson, C. R., & Siegel, A. F. (1987). Parsimonious modeling of yield curves. *The Journal of Business*, *60*(4), 473–489. https://www.jstor.org/stable/2352957

Patel, K., Mohamed, A., & Van Wuuren, G. W. (2018). A regression and comparative study of United States and South African yield curves using principal component analysis. *South African Journal of Economic and Management Sciences*, *21*(1), Article a1626, 791–819. https://doi.org/10.4102/sajems.v21i1.1626

Piazzesi, M. (2010). Affine term structure models. In Y. Aït-Sahalia & L. P. Hansen (Eds.), *Handbook of financial econometrics: Tools and techniques* (pp. 691–766, Vol. 1). North-Holland. https://doi.org/10.1016/B978-0-444-50897-3.50015-8

Rigobon, R., & Sack, B. (2003). Measuring the reaction of monetary policy to the stock market. *The Quarterly Journal of Economics*, *118*(2), 639–669. https://www.jstor.org/stable/25053916

Romer, C. D., & Romer, D. H. (2000). Federal Reserve information and the behavior of interest rates. *The American Economic Review*, *90*(3), 429–457. https://doi.org/10.1257/aer.90.3.429

Swanson, E. T. (2006). Have increases in Federal Reserve transparency improved private sector interest rate forecasts? *Journal of Money, Credit and Banking*, *38*(3), 791–819. https://muse.jhu.edu/article/198106/pdf

Swanson, E. T. (2011). *Let's twist again: A high-frequency event-study analysis of operation twist and its implications for QE2* (Working Paper No. 2011-08) (Version 1.0, February 19, 2011). Federal Reserve Bank of San Francisco. http://www.frbsf.org/publications/economics/papers/2011/wp11-08bk.pdf

Swanson, E. T., & Jayawickrema, V. (2024). *Speeches by the Fed Chair are more important than FOMC announcements: An improved high-frequency measure of U.S. monetary policy shocks* (Working Paper) (Version 2.0, March 27, 2024). https://sites.socsci.uci.edu/~swanson2/papers/hfdat.pdf

Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage.

Appendix

Table A.1: Extended Specification with Conservative (HC3) Standard Errors

	Level	Slope	Curvature
$mps^{\perp}$	1.034***	0.282***	-0.041**
	(0.165)	(0.044)	(0.020)
D^{info}	0.007	-0.007*	-0.001
	(0.009)	(0.004)	(0.002)
$mps^{\perp} \times D^{info}$	0.155	-0.134*	-0.020
	(0.230)	(0.072)	(0.030)
Observations	192	192	192
R^2	0.431	0.199	0.039

Notes: The table reports regression results of the first three principal components of Treasury yield changes on orthogonal MPS, an indicator for information-dominated events, and their interaction term. Thus, columns correspond to changes in the first, second, and third principal component. Robust standard errors (HC3) are reported in parentheses, and statistical significance is based on corresponding robust p-values. *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A.2: Extended Specification with Sample Period 1995-2023

	Level	Slope	Curvature
$mps^{\perp}$	1.126***	0.224***	-0.032**
	(0.118)	(0.038)	(0.015)
D^{info}	0.006	-0.006*	-0.000
	(0.007)	(0.003)	(0.002)
$mps^{\perp} \times D^{info}$	0.019	-0.111*	-0.015
	(0.181)	(0.059)	(0.023)
Observations	233	233	233
R^2	0.521	0.166	0.029

Notes: The table reports regression results of the first three principal components of Treasury yield changes on orthogonal MPS, an indicator for information-dominated events, and their interaction term. Thus, columns correspond to changes in the first, second, and third principal component. The sample period spans from February 1995 to December 2023. Robust standard errors (HC1) are reported in parentheses. Statistical significance is based on corresponding robust p-values. *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A.3: Extended Specification with Sample Period 1997-2023

	Level	Slope	Curvature
$mps^{\perp}$	1.099***	0.252***	-0.032**
	(0.133)	(0.037)	(0.016)
D^{info}	0.007	-0.006*	-0.000
	(0.008)	(0.004)	(0.002)
$mps^{\perp} \times D^{info}$	0.086	-0.136**	-0.027
	(0.194)	(0.061)	(0.025)
Observations	212	212	212
R^2	0.488	0.183	0.031

Notes: The table reports regression results of the first three principal components of Treasury yield changes on orthogonal MPS, an indicator for information-dominated events, and their interaction term. Thus, columns correspond to changes in the first, second, and third principal component. The sample period spans from September 1997 to December 2023. Robust standard errors (HC1) are reported in parentheses. Statistical significance is based on corresponding robust p-values. *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A.4: Extended Specification with Conventional MPS

	Level	Slope	Curvature
$mps^{conventional}$	0.984***	0.242***	-0.033
	(0.133)	(0.042)	(0.022)
D^{info}	-0.002	-0.006	-0.001
	(0.007)	(0.004)	(0.002)
$mps^{conventional} \times D^{info}$	0.540***	-0.135*	-0.024
	(0.166)	(0.082)	(0.030)
Observations	204	204	204
R^2	0.567	0.189	0.038

Notes: The table reports regression results of the first three principal components of Treasury yield changes on conventional MPS, an indicator for information-dominated events, and their interaction term. Thus, columns correspond to changes in the first, second, and third principal component. Robust standard errors (HC1) are reported in parentheses. Statistical significance is based on corresponding robust p-values. *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A.5: Extended Specification with Stricter Event Classification

	Level	Slope	Curvature
$mps^{\perp}$	1.029***	0.280***	-0.042**
	(0.162)	(0.044)	(0.020)
D^{info}	0.004	-0.008*	-0.001
	(0.010)	(0.005)	(0.003)
$mps^{\perp} \times D^{info}$	0.152	-0.129*	-0.024
	(0.225)	(0.073)	(0.032)
Observations	155	155	155
R^2	0.435	0.203	0.043

Notes: The table reports regression results of the first three principal components of Treasury yield changes on orthogonal MPS, an indicator for information-dominated events, and their interaction term. Thus, columns correspond to changes in the first, second, and third principal component. For the classification of event types, the ten percent of events with the stock response, the MPS, or both closest to zero are excluded. Robust standard errors (HC1) are reported in parentheses. Statistical significance is reported based on corresponding robust p-values and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A.6: Extended Specification with Treasury Yield Spreads

	30-2 year spread	30-5 year spread	10-2 year spread
$mps^{\perp}$	-0.588***	-0.464***	-0.369***
	(0.053)	(0.046)	(0.044)
D^{info}	0.007	0.005	0.005
	(0.005)	(0.005)	(0.003)
$mps^{\perp} \times D^{info}$	0.160*	0.091	0.074
	(0.093)	(0.086)	(0.066)
Observations	192	192	192
R^2	0.434	0.397	0.345

Notes: The table reports regression results of spreads between Treasuries of different maturities on orthogonal MPS, an indicator for information-dominated events, and their interaction term. Thus, columns correspond to changes in the spreads between the 30-year and 2-year, 30-year and 5-year, and 10-year and 2-year Treasury. Robust standard errors (HC1) are reported in parentheses. Statistical significance is based on corresponding robust p-values and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Declaration of Authorship

”I hereby declare,

- that I have written this thesis independently;
- that I have written the thesis using only the aids specified in the index;
- that all parts of the thesis produced with the help of aids have been declared;
- that I have handled both input and output responsibly when using AI. I confirm that I have therefore only read in public data or data released with consent and that I have checked, declared and comprehensibly referenced all results and/or other forms of AI assistance in the required form and that I am aware that I am responsible if incorrect content, violations of data protection law, copyright law or scientific misconduct (e.g. plagiarism) have also occurred unintentionally;
- that I have mentioned all sources used and cited them correctly according to established academic citation rules;
- that I have acquired all immaterial rights to any materials I may have used, such as images or graphics, or that these materials were created by me;
- that the topic, the thesis or parts of it have not already been the object of any work or examination of another course, unless this has been expressly agreed with the faculty member in advance and is stated as such in the thesis;
- that I am aware of the legal provisions regarding the publication and dissemination of parts or the entire thesis and that I comply with them accordingly;
- that I am aware that my thesis can be electronically checked for plagiarism and for third-party authorship of human or technical origin and that I hereby grant the University of St. Gallen the copyright according to the Examination Regulations as far as it is necessary for the administrative actions;
- that I am aware that the University will prosecute a violation of this Declaration of Authorship and that disciplinary as well as criminal consequences may result, which may lead to expulsion from the University or to the withdrawal of my title.”

By submitting this thesis, I confirm through my conclusive action that I am submitting the Declaration of Authorship, that I have read and understood it, and that it is true.

19 May 2026, St. Gallen **Signature:** 

Directory of Auxiliary Aids

Table A.7: Use of Supplementary Materials

Resource	**Usage**	**Affected Sections**
Literature	As indicated in the references	Entire thesis
ChatGPT	Writing: used for refinements of structure and style	Entire thesis
ChatGPT	Coding: used to debug and restructure code	Code and Tables
ChatGPT	Formatting: used for correct formatting and export in LaTeX	Entire thesis
DeepL	Translation, spelling and grammar	Entire thesis